

PRACTICAL NO. 1

Aim: Implement IP SLA (IP Service Level Agreement)

Theory :

IP SLA is a networking feature that monitors real-time performance metrics like latency, jitter, packet loss, and path availability, ensuring efficient network operations.

Its goal is to proactively detect issues, optimize application performance, and ensure SLA compliance by simulating real-time traffic.

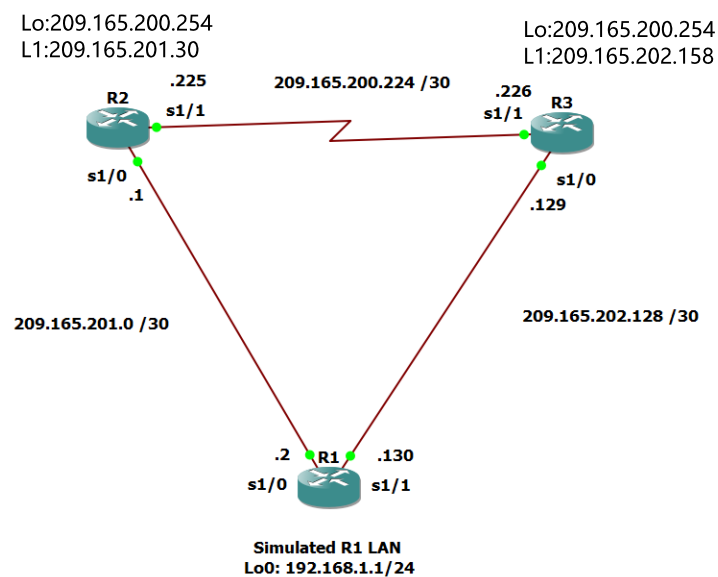
Key Components :

- **Operation Types:** ICMP Echo, HTTP, FTP, VoIP, DNS.
- **Metrics:** Round-trip time (RTT), jitter, response time, and packet loss.

Practical Application :

- IP SLA helps monitor critical paths, identify bottlenecks, and ensure seamless end-to-end connectivity for applications.

Topology:



Step 1: Configure loopbacks and assign addresses.

Router R1

```
config t
```

```
hostname R1
```

```
int loopback 0
```

```
desc R1 LAN
```

```
ip add 192.168.1.1 255.255.255.0
```

```
R1#config t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#hostname R1
R1(config)#int loopback 0
R1(config-if)#
*Sep 13 18:07:15.415: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
R1(config-if)#desc R1 LAN
R1(config-if)#ip add 192.168.1.1 255.255.255.0
```

int s1/0

desc R1 --> ISP1

ip add 209.165.201.2 255.255.255.252

clock rate 128000

bandwidth 128

no sh

```
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int s1/0
R1(config-if)#desc R1 --> ISP1
R1(config-if)#ip add 209.165.201.2 255.255.255.252
R1(config-if)#clock rate 128000
R1(config-if)#bandwidth 128
R1(config-if)#no sh
R1(config-if)#
*Sep 13 18:12:05.147: %LINK-3-UPDOWN: Interface Serial1/0, changed state to up
R1(config-if)#
*Sep 13 18:12:06.151: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0, changed state to up
R1(config-if)#
*Sep 13 18:12:28.199: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0, changed state to down
```

int s1/1

ip add 209.165.202.130 255.255.255.252

bandwidth 128

no sh

```
R1(config-if)#int s1/1
R1(config-if)#ip add 209.165.202.130 255.255.255.252
R1(config-if)#bandwidth 128
R1(config-if)#no sh
R1(config-if)#
*Sep 13 18:14:46.039: %LINK-3-UPDOWN: Interface Serial1/1, changed state to up
R1(config-if)#
*Sep 13 18:14:47.047: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/1, changed state to up
R1(config-if)#
R1#
*Sep 13 18:14:54.083: %SYS-5-CONFIG_I: Configured from console by console
R1#
```

Router R2

hostname ISP1

interface Loopback0

description Simulated Internet Web Server

ip address 209.165.200.254 255.255.255.255

interface Loopback1

description ISP1 DNS Server

ip address 209.165.201.30 255.255.255.255

```
interface Serial1/0
description ISP1 --> R1
ip address 209.165.201.1 255.255.255.252
bandwidth 128

no shutdown

interface Serial1/1
description ISP1 --> ISP2
ip address 209.165.200.225 255.255.255.252
clock rate 128000
bandwidth 128

no shutdown
```

```
ISP1(config-if)#interface Loopback0
ISP1(config-if)#description Simulated Internet Web Server
ISP1(config-if)#ip address 209.165.200.254 255.255.255.255
ISP1(config-if)#interface Loopback1
ISP1(config-if)#description ISP1 DNS Server
ISP1(config-if)#ip address 209.165.201.30 255.255.255.255
ISP1(config-if)#interface Serial1/0
ISP1(config-if)#description ISP1 --> R1
ISP1(config-if)#ip address 209.165.201.1 255.255.255.252
ISP1(config-if)#bandwidth 128
ISP1(config-if)#no shutdown
ISP1(config-if)#
*Sep 13 18:35:00.859: %LINK-3-UPDOWN: Interface Serial1/0, changed state to up
ISP1(config-if)#
*Sep 13 18:35:01.867: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0, changed state to up
ISP1(config-if)#interface Serial1/1
ISP1(config-if)#description ISP1 --> ISP2
ISP1(config-if)#ip address 209.165.200.225 255.255.255.252
ISP1(config-if)#clock rate 128000
ISP1(config-if)#bandwidth 128
ISP1(config-if)#no shutdown
ISP1(config-if)#
*Sep 13 18:35:59.347: %LINK-3-UPDOWN: Interface Serial1/1, changed state to up
ISP1(config-if)#
*Sep 13 18:36:00.355: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/1, changed state to up
ISP1(config-if)#
*Sep 13 18:36:24.219: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/1, changed state to down
ISP1(config-if)#
```

Router ISP2 (R3)

```
hostname ISP2

interface Loopback0
description Simulated Internet Web Server
ip address 209.165.200.254 255.255.255.255

interface Loopback1
description ISP2 DNS Server
ip address 209.165.202.158 255.255.255.255

interface Serial1/0
description ISP2 --> R1
```

```
ip address 209.165.202.129 255.255.255.252
```

```
clock rate 128000
```

```
bandwidth 128
```

```
no shutdown
```

```
interface Serial1/1
```

```
description ISP2 --> ISP1
```

```
ip address 209.165.200.226 255.255.255.252
```

```
bandwidth 128
```

```
no shutdown
```

```
R3#config
Configuring from terminal, memory, or network [terminal]? t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#hostname ISP2
ISP2(config)#interface Loopback0
ISP2(config-if)#
*Sep 13 18:37:30.255: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
ISP2(config-if)#description Simulated Internet Web Server
ISP2(config-if)#ip address 209.165.200.254 255.255.255.255
ISP2(config-if)#interface Loopback1
ISP2(config-if)#
*Sep 13 18:38:11.579: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback1, changed state to up
ISP2(config-if)#description ISP2 DNS Server
ISP2(config-if)#ip address 209.165.202.158 255.255.255.255
ISP2(config-if)#interface Serial1/0
ISP2(config-if)#description ISP2 --> R1
ISP2(config-if)#ip address 209.165.202.129 255.255.255.252
ISP2(config-if)#clock rate 128000
ISP2(config-if)#bandwidth 128
ISP2(config-if)#no shutdown
ISP2(config-if)#
*Sep 13 18:40:24.083: %LINK-3-UPDOWN: Interface Serial1/0, changed state to up
ISP2(config-if)#
*Sep 13 18:40:25.095: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0, changed state to up
ISP2(config-if)#interface Serial1/1
ISP2(config-if)#description ISP2 --> ISP1
ISP2(config-if)#ip address 209.165.200.226 255.255.255.252
ISP2(config-if)#bandwidth 128
ISP2(config-if)#no shutdown
ISP2(config-if)#
*Sep 13 18:41:38.467: %LINK-3-UPDOWN: Interface Serial1/1, changed state to up
ISP2(config-if)#
*Sep 13 18:41:39.475: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/1, changed state to up
ISP2(config-if)#
```

Verify the configuration by using the show interfaces description command. The output from router R1 is shown here as an example.

```
R1# show interfaces description | include up
```

```
R1#show interfaces description | include up
Se1/0                                up                up                R1 --> ISP1
Se1/1                                up                up                R1 --> ISP1
Lo0                                  up                up                R1 LAN
R1#sh ip int br
```

Step 2: Configure static routing

Router R1

```
R1(config)# ip route 0.0.0.0 0.0.0.0 209.165.201.1
```

```
R1(config)#ip route 0.0.0.0 0.0.0.0 209.165.201.1
R1(config)#
```

Router ISP1 (R2)

conf t

router eigrp 1

network 209.165.200.224 0.0.0.3

network 209.165.201.0 0.0.0.31

no auto-summary

exit

ip route 192.168.1.0 255.255.255.0 209.165.201.2

end

wr

```
SP1(config)#router eigrp 1
SP1(config-router)#network 209.165.200.224 0.0.0.3
SP1(config-router)#network 209.165.201.0 0.0.0.31
SP1(config-router)#no auto-summary
SP1(config-router)#exit
SP1(config)#ip route 192.168.1.0 255.255.255.0 209.165.201.2
SP1(config)#
Sep 13 18:47:49.331: %DUAL-5-NBRCHANGE: EIGRP-IPv4 1: Neighbor 209.165.200.226 (Serial1/1) is up: new adjacency
SP1(config)#
```

Router ISP2 (R3)

conf t

router eigrp 1

network 209.165.200.224 0.0.0.3

network 209.165.202.128 0.0.0.31

no auto-summary

exit

ip route 192.168.1.0 255.255.255.0 209.165.202.130

end

wr

```
ISP2(config)#router eigrp 1
ISP2(config-router)#network 209.165.200.224 0.0.0.3
ISP2(config-router)#
*Sep 13 18:47:49.391: %DUAL-5-NBRCHANGE: EIGRP-IPv4 1: Neighbor 209.165.200.225 (Serial1/1) is up: new adjace
ISP2(config-router)#network 209.165.202.128 0.0.0.31
ISP2(config-router)#no auto-summary
ISP2(config-router)#exit
ISP2(config)#
ISP2(config)#ip route 192.168.1.0 255.255.255.0 209.165.202.130
ISP2(config)#
```

Before implementing the Cisco IOS SLA feature, you must verify reachability to the Internet servers. From router R1, ping the web server, ISP1 DNS server, and ISP2 DNS server to verify connectivity. You can copy the following Tcl script and paste it into R1.

```
tclsh

foreach address {
209.165.200.254
209.165.201.30
209.165.202.158
} {
ping $address source 192.168.1.1
}
}
```

```
R1#tclsh
R1(tcl)#foreach address {
+>(tcl)#209.165.200.254
+>(tcl)#209.165.201.30
+>(tcl)#209.165.202.158
+>(tcl)#} {
+>(tcl)#ping $address source 192.168.1.1
+>(tcl)#}
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 209.165.200.254, timeout is 2 seconds:
Packet sent with a source address of 192.168.1.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 24/30/40 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 209.165.201.30, timeout is 2 seconds:
Packet sent with a source address of 192.168.1.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 24/28/36 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 209.165.202.158, timeout is 2 seconds:
Packet sent with a source address of 192.168.1.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 36/46/52 ms
```

Trace the path taken to the web server, ISP1 DNS server, and ISP2 DNS server. You can copy the following Tcl script and paste it into R1.

```
tclsh

foreach address {
209.165.200.254
209.165.201.30
209.165.202.158
} {
trace $address source 192.168.1.1
}
```

}

```
R1(tcl)#foreach address {
+>(tcl)#209.165.200.254
+>(tcl)#209.165.201.30
+>(tcl)#209.165.202.158
+>(tcl)#} {
+>(tcl)#trace $address source 192.168.1.1
+>(tcl)#}
Type escape sequence to abort.
Tracing the route to 209.165.200.254
VRF info: (vrf in name/id, vrf out name/id)
  1 209.165.201.1 28 msec 32 msec 32 msec
Type escape sequence to abort.
Tracing the route to 209.165.201.30
VRF info: (vrf in name/id, vrf out name/id)
  1 209.165.201.1 20 msec 28 msec 28 msec
Type escape sequence to abort.
Tracing the route to 209.165.202.158
VRF info: (vrf in name/id, vrf out name/id)
  1 209.165.201.1 32 msec 32 msec 28 msec
  2 209.165.200.226 40 msec 48 msec 48 msec
R1(tcl)#
```

Step 3: Configure IP SLA probes.

- a. Create an ICMP echo probe on R1 to the primary DNS server on ISP1 using the ip sla command.

conf t

ip sla 11

icmp-echo 209.165.201.30

frequency 10

exit

ip sla schedule 11 life forever start-time now

end

wr

```
R1#config t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ip sla 11
R1(config-ip-sla)#icmp-echo 209.165.201.30
R1(config-ip-sla-echo)#frequency 10
R1(config-ip-sla-echo)#exit
R1(config)#ip sla schedule 11 life forever start-time now
R1(config)#
```

- b. Verify the IP SLAs configuration of operation 11 using the show ip sla configuration 11 command.

R1# show ip sla configuration 11

```

R1#show ip sla configuration 11
IP SLAs Infrastructure Engine-III
Entry number: 11
Owner:
Tag:
Operation timeout (milliseconds): 5000
Type of operation to perform: icmp-echo
Target address/Source address: 209.165.201.30/0.0.0.0
Type Of Service parameter: 0x0
Request size (ARR data portion): 28
Verify data: No
Vrf Name:
Schedule:
  Operation frequency (seconds): 10 (not considered if randomly scheduled)
  Next Scheduled Start Time: Start Time already passed
  Group Scheduled : FALSE
  Randomly Scheduled : FALSE
  Life (seconds): Forever
  Entry Ageout (seconds): never
  Recurring (Starting Everyday): FALSE
  Status of entry (SNMP RowStatus): Active
Threshold (milliseconds): 5000
Distribution Statistics:
  Number of statistic hours kept: 2
  Number of statistic distribution buckets kept: 1
  Statistic distribution interval (milliseconds): 20
Enhanced History:
History Statistics:
  Number of history Lives kept: 0
  Number of history Buckets kept: 15
  History Filter Type: None

```

c. Issue the show ip sla statistics command to display the number of successes, failures, and results of the latest operations.

R1# show ip sla statistics

```

R1#show ip sla statistics
IPSLAs Latest Operation Statistics

IPSLA operation id: 11
  Latest RTT: 32 milliseconds
Latest operation start time: 19:55:13 UTC Sun Oct 5 2025
Latest operation return code: OK
Number of successes: 4
Number of failures: 0
Operation time to live: Forever

```

You can see that operation 11 has already succeeded five times, has had no failures, and the last operation returned an OK result.

d. Although not actually required because IP SLA session 11 alone could provide the desired fault tolerance, create a second probe, 22, to test connectivity to the second DNS server located on router ISP2.

conf t

ip sla 22

icmp-echo 209.165.202.158

frequency 10

exit

ip sla schedule 22 life forever start-time now

end

WR

```
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)# ip sla 22
R1(config-ip-sla)# icmp-echo 209.165.202.158
R1(config-ip-sla-echo)# frequency 10
R1(config-ip-sla-echo)# exit
R1(config)#
R1(config)# ip sla schedule 22 life forever start-time now
R1(config)#end
R1#
```

e. Verify the new probe using the show ip sla configuration and show ip sla statistics commands.

R1# show ip sla configuration 22

```
R1#show ip sla configuration 22
IP SLAs Infrastructure Engine-III
Entry number: 22
Owner:
Tag:
Operation timeout (milliseconds): 5000
Type of operation to perform: icmp-echo
Target address/Source address: 209.165.202.158/0.0.0.0
Type Of Service parameter: 0x0
Request size (ARR data portion): 28
Verify data: No
Vrf Name:
Schedule:
  Operation frequency (seconds): 10 (not considered if randomly scheduled)
  Next Scheduled Start Time: Start Time already passed
  Group Scheduled : FALSE
  Randomly Scheduled : FALSE
  Life (seconds): Forever
  Entry Ageout (seconds): never
  Recurring (Starting Everyday): FALSE
  Status of entry (SNMP RowStatus): Active
Threshold (milliseconds): 5000
Distribution Statistics:
  Number of statistic hours kept: 2
  Number of statistic distribution buckets kept: 1
  Statistic distribution interval (milliseconds): 20
Enhanced History:
History Statistics:
  Number of history Lives kept: 0
  Number of history Buckets kept: 15
History Filter Type: None
```

R1# show ip sla statistics 22

```
R1#show ip sla statistics 22
IPSLAs Latest Operation Statistics

IPSLA operation id: 22
  Latest RTT: 64 milliseconds
Latest operation start time: 19:57:47 UTC Sun Oct 5 2025
Latest operation return code: OK
Number of successes: 7
Number of failures: 0
Operation time to live: Forever
```

The output lists the details of the configuration of operation 22. The operation is an ICMP echo to 209.165.202.158, with a frequency of 10 seconds, and it has already started (the start time has already

passed). The statistics also prove that operation 22 is active.

Step 4: Configure tracking options.

Although PBR could be used, you will configure a floating static route that appears or disappears depending

on the success or failure of the IP SLA.

a. On R1, remove the current default route and replace it with a floating static route having an administrative

distance of 5.

```
R1(config)# no ip route 0.0.0.0 0.0.0.0 209.165.201.1
```

```
R1(config)# ip route 0.0.0.0 0.0.0.0 209.165.201.1 5
```

```
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#no ip route 0.0.0.0 0.0.0.0 209.165.201.1
R1(config)#ip route 0.0.0.0 0.0.0.0 209.165.201.1 5
R1(config)#
```

b. Verify the routing table.

```
R1# show ip route | begin Gateway
```

```
R1#show ip route | begin Gateway
Gateway of last resort is 209.165.201.1 to network 0.0.0.0

S*    0.0.0.0/0 [5/0] via 209.165.201.1
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.1.0/24 is directly connected, Loopback0
L      192.168.1.1/32 is directly connected, Loopback0
      209.165.201.0/24 is variably subnetted, 2 subnets, 2 masks
C      209.165.201.0/30 is directly connected, Serial1/0
L      209.165.201.2/32 is directly connected, Serial1/0
      209.165.202.0/24 is variably subnetted, 2 subnets, 2 masks
C      209.165.202.128/30 is directly connected, Serial1/1
L      209.165.202.130/32 is directly connected, Serial1/1
R1#
```

Notice that the default static route is now using the route with the administrative distance of 5. The first

tracking object is tied to IP SLA object 11.

c. From global configuration mode on R1, use the track 1 ip sla 11 reachability command to enter the config-track subconfiguration mode.

```
R1(config)# track 1 ip sla 11 reachability
```

```
Enter configuration commands, one per line.
R1(config)#track 1 ip sla 11 reachability
R1(config-track)#
```

d. Specify the level of sensitivity to changes of tracked objects to 10 seconds of down delay and 1 second of up delay using the delay down 10 up 1 command. The delay helps to alleviate the effect of flapping objects—objects that are going down and up rapidly. In this situation, if the DNS server fails momentarily and comes back up within 10 seconds, there is no impact.

```
R1(config-track)# delay down 10 up 1
```

```
R1(config-track)# exit
```

```
R1(config)#
```

```
R1(config)#track 1 ip sla 11 reachability
R1(config-track)#delay down 10 up 1
R1(config-track)#exit
R1(config)#
```

e. To view routing table changes as they happen, first enable the debug ip routing command.

```
R1# debug ip routing
```

```
R1#debug ip routing
IP routing debugging is on
R1#
```

f. Configure the floating static route that will be implemented when tracking object 1 is active. Use the ip route 0.0.0.0 0.0.0.0 209.165.201.1 2 track 1 command to create a floating static default route via 209.165.201.1 (ISP1). Notice that this command references the tracking object number 1, which in turn references IP SLA operation number 11.

```
R1(config)# ip route 0.0.0.0 0.0.0.0 209.165.201.1 2 track 1
```

```
R1(config)#ip route 0.0.0.0 0.0.0.0 209.165.201.1 2 track 1
R1(config)#
*Oct  5 20:06:51.831: RT: updating static 0.0.0.0/0 (0x0):
      via 209.165.201.1    1048578

*Oct  5 20:06:51.835: RT: closer admin distance for 0.0.0.0, flushing 1 routes
*Oct  5 20:06:51.835: RT: add 0.0.0.0/0 via 209.165.201.1, static metric [2/0]
*Oct  5 20:06:51.839: RT: updating static 0.0.0.0/0 (0x0):
      via 209.165.201.1    1048578

*Oct  5 20:06:51.843: RT: rib update return code: 17
*Oct  5 20:06:51.843: RT: updating static 0.0.0.0/0 (0x0):
      via 209.165.201.1    1048578

*Oct  5 20:06:51.843: RT: rib update return code: 17
R1(config)#
```

Notice that the default route with an administrative distance of 5 has been immediately flushed because of a route with a better admin distance. It then adds the new default route with the admin distance of 2.

g. Repeat the steps for operation 22, track number 2, and assign the static route an admin distance higher than track 1 and lower than 5. On R1, copy the following configuration, which sets an admin distance of 3.

track 2 ip sla 22 reachability

delay down 10 up 1

exit

ip route 0.0.0.0 0.0.0.0 209.165.202.129 3 track 2

```
R1(config)#track 2 ip sla 22 reachability
R1(config-track)# delay down 10 up 1
R1(config-track)#exit
R1(config)#ip route 0.0.0.0 0.0.0.0 209.165.202.129 3 track 2
R1(config)#
*Oct  5 20:08:35.515: RT: updating static 0.0.0.0/0 (0x0):
      via 209.165.201.1    1048578

*Oct  5 20:08:35.519: RT: updating static 0.0.0.0/0 (0x0):
      via 209.165.201.1    1048578

*Oct  5 20:08:35.523: RT: rib update return code: 17
*Oct  5 20:08:35.531: RT: updating static 0.0.0.0/0 (0x0):
      via 209.165.202.129  1048578

*Oct  5 20:08:35.531: RT: rib update return code: 17
```

h. Verify the routing table again.

R1#show ip route | begin Gateway

```
R1#show ip route | begin Gateway
Gateway of last resort is 209.165.201.1 to network 0.0.0.0

S*    0.0.0.0/0 [2/0] via 209.165.201.1
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.1.0/24 is directly connected, Loopback0
L      192.168.1.1/32 is directly connected, Loopback0
      209.165.201.0/24 is variably subnetted, 2 subnets, 2 masks
C      209.165.201.0/30 is directly connected, Serial1/0
L      209.165.201.2/32 is directly connected, Serial1/0
      209.165.202.0/24 is variably subnetted, 2 subnets, 2 masks
C      209.165.202.128/30 is directly connected, Serial1/1
L      209.165.202.130/32 is directly connected, Serial1/1
R1#
*Oct  5 20:08:55.439: %SYS-5-CONFIG_I: Configured from console by console
R1#
```

Although a new default route was entered, its administrative distance is not better than 2. Therefore, it does not replace the previously entered default route.

Step 5: Verify IP SLA operation.

In this step you observe and verify the dynamic operations and routing changes when tracked objects fail.

The following summarizes the process:

- Disable the DNS loopback interface on ISP1 (R2).
- Observe the output of the debug command on R1.
- Verify the static route entries in the routing table and the IP SLA statistics of R1.
- Re-enable the loopback interface on ISP1 (R2) and again observe the operation of the IP SLA tracking feature.

a. On ISP1, disable the loopback interface 1.

R2:

```
int lo1
shutdown
```

```
ISP1(config)#int lo1
ISP1(config-if)#shutdown
ISP1(config-if)#
```

b. On R1, observe the debug output being generated. Recall that R1 will wait up to 10 seconds before initiating action therefore several seconds will elapse before the output is generated.

```
*Oct  5 20:37:53.831: %TRACKING-5-STATE: 1 ip sla 11 reachability Up->Down
*Oct  5 20:37:53.835: RT: del 0.0.0.0 via 209.165.201.1, static metric [2/0]
*Oct  5 20:37:53.835: RT: delete network route to 0.0.0.0/0
*Oct  5 20:37:53.835: RT: default path has been cleared
*Oct  5 20:37:53.835: RT: updating static 0.0.0.0/0 (0x0):
      via 209.165.202.129  1048578

*Oct  5 20:37:53.835: RT: add 0.0.0.0/0 via 209.165.202.129, static metric [3/0]
*Oct  5 20:37:53.835: RT: default path is now 0.0.0.0 via 209.165.202.129
*Oct  5 20:37:53.835: RT: updating static 0.0.0.0/0 (0x0):
      via 209.165.201.1  1048578

*Oct  5 20:37:53.835: RT: rib update return code: 17
*Oct  5 20:37:53.847: RT: updating static 0.0.0.0/0 (0x0):
      via 209.165.202.129  1048578

R1#
*Oct  5 20:37:53.851: RT: updating static 0.0.0.0/0 (0x0):
      via 209.165.201.1  1048578

*Oct  5 20:37:53.851: RT: rib update return code: 17
```

The tracking state of track 1 changes from up to down. This is the object that tracked reachability for IP

SLA object 11, with an ICMP echo to the ISP1 DNS server at 209.165.201.30.

R1 then proceeds to delete the default route with the administrative distance of 2 and installs the next highest default route to ISP2 with the administrative distance of 3.

c. On R1, verify the routing table.

R1# show ip route | begin Gateway

```

R1#show ip route | begin Gateway
Gateway of last resort is 209.165.202.129 to network 0.0.0.0

S*    0.0.0.0/0 [3/0] via 209.165.202.129
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.1.0/24 is directly connected, Loopback0
L      192.168.1.1/32 is directly connected, Loopback0
      209.165.201.0/24 is variably subnetted, 2 subnets, 2 masks
C      209.165.201.0/30 is directly connected, Serial1/0
L      209.165.201.2/32 is directly connected, Serial1/0
      209.165.202.0/24 is variably subnetted, 2 subnets, 2 masks
C      209.165.202.128/30 is directly connected, Serial1/1
L      209.165.202.130/32 is directly connected, Serial1/1

```

d. Verify the IP SLA statistics.

```
R1# show ip sla statistics
```

```

R1#show ip sla statistics
IPSLAs Latest Operation Statistics

```

```

IPSLA operation id: 11
    Latest RTT: NoConnection/Busy/Timeout
Latest operation start time: 20:39:13 UTC Sun Oct 5 2025
Latest operation return code: Timeout
Number of successes: 257
Number of failures: 11
Operation time to live: Forever

```

```

IPSLA operation id: 22
    Latest RTT: 32 milliseconds
Latest operation start time: 20:39:17 UTC Sun Oct 5 2025
Latest operation return code: OK
Number of successes: 255
Number of failures: 1
Operation time to live: Forever

```

e. On R1, initiate a trace to the web server from the internal LAN IP address.

```
R1# trace 209.165.200.254 source 192.168.1.1
```

```

R1#trace 209.165.200.254 source 192.168.1.1
Type escape sequence to abort.
Tracing the route to 209.165.200.254
VRF info: (vrf in name/id, vrf out name/id)
  1 209.165.201.1 32 msec 28 msec 24 msec

```

f. On ISP1, re-enable the DNS address by issuing the no shutdown command on the loopback 1 interface to examine the routing behavior when connectivity to the ISP1 DNS is restored.

```
ISP1(config-if)# no shutdown
```

```
ISP1(config-if)#no shutdown
ISP1(config-if)#
```

g. Again examine the IP SLA statistics.

R1# show ip sla statistics

```
R1#show ip sla statistics
IPSLAs Latest Operation Statistics

IPSLA operation id: 11
    Latest RTT: 20 milliseconds
Latest operation start time: 20:40:43 UTC Sun Oct 5 2025
Latest operation return code: OK
Number of successes: 260
Number of failures: 17
Operation time to live: Forever

IPSLA operation id: 22
    Latest RTT: 76 milliseconds
Latest operation start time: 20:40:37 UTC Sun Oct 5 2025
Latest operation return code: OK
Number of successes: 263
Number of failures: 1
Operation time to live: Forever
```

h. Verify the routing table.

R1# show ip route | begin Gateway

```
R1#show ip route | begin Gateway
Gateway of last resort is 209.165.201.1 to network 0.0.0.0

S*    0.0.0.0/0 [2/0] via 209.165.201.1
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.1.0/24 is directly connected, Loopback0
L      192.168.1.1/32 is directly connected, Loopback0
      209.165.201.0/24 is variably subnetted, 2 subnets, 2 masks
C      209.165.201.0/30 is directly connected, Serial1/0
L      209.165.201.2/32 is directly connected, Serial1/0
      209.165.202.0/24 is variably subnetted, 2 subnets, 2 masks
C      209.165.202.128/30 is directly connected, Serial1/1
L      209.165.202.130/32 is directly connected, Serial1/1
```